

DATA STRUCTURE

LAB PROGRAM

Write a C program to implement Heap sort

```
#include <stdio.h>
```

```
void heapify(int a[], int n, int i)
```

```
{
```

```
    int largest = i;
```

```
    int left = 2 * i + 1;
```

```
    int right = 2 * i + 2;
```

```
    int temp;
```

```
    if(left < n && a[left] > a[largest])
```

```
        largest = left;
```

```
    if(right < n && a[right] > a[largest])
```

```
        largest = right;
```

```
    if(largest != i)
```

```
{
```

```
        temp = a[i];
```

```
        a[i] = a[largest];
```

```
        a[largest] = temp;
```

```
        heapify(a, n, largest);
```

```
}
```

```
}
```

```
void heapsort(int a[], int n)
```

```
{
```

```
    int i, temp;
```

```
    for(i = n / 2 - 1; i >= 0; i--)
```

```
        heapify(a, n, i);
```

```
    for(i = n - 1; i > 0; i--)
```

```
    {
```

```
        temp = a[0];
```

```
        a[0] = a[i];
```

```
        a[i] = temp;
```

```
        heapify(a, i, 0);
```

```
    }
```

```
}
```

```
int main()
```

```
{
```

```
    int a[100], n, i;
```

```
    printf("Enter number of elements: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter elements:\n");
```

```
    for(i = 0; i < n; i++)
```

```
scanf("%d", &a[i]);

heapsort(a, n);

printf("Sorted array:\n");
for(i = 0; i < n; i++)
    printf("%d ", a[i]);

return 0;
}
```

Output

```
Enter number of elements: 2
Enter elements:
97 5
Sorted array:
5 97
```

=== Code Execution Successful ===